

# 5DGSPTSBL: Geometric Fast-Slow Stable Boundary Layer Model

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## Abstract

We present a five-dimensional geometric singular perturbation model for the stable boundary layer with state variables  $(\tilde{e}, q_\theta, S, T_s, T_g)$ . The model combines desingularized turbulence dynamics, slaved thermodynamic closures, and slow thermal coupling to support fold-locus diagnostics and data-driven parameter identification.

## 1 Closures and Smooth Regularization

To maintain  $C^r$ -smoothness near neutral stratification and support robust implicit integration with autodiff Jacobians, the stratification closure uses a hyperbolic embedding of  $\max(0, x)$ :

$$\text{smooth\_max}(x; \epsilon) = \frac{x + \sqrt{x^2 + \epsilon^2}}{2}. \quad (1)$$

With  $x = (\theta_0 - T_s)/h_{\text{sbl}}(\tilde{e})$ , the closure becomes

$$\theta_z(T_s, \tilde{e}) = \text{smooth\_max}\left(\frac{\theta_0 - T_s}{h_{\text{sbl}}(\tilde{e})}; \epsilon_{\text{strat}}\right), \quad (2)$$

where  $\epsilon_{\text{strat}}$  is a small smoothing scale exposed in `SBLParams`.

The bulk transfer coefficients are

$$C_D = \left[ \frac{\kappa}{\ln((z_1 - d)/z_{0m})} \right]^2, \quad (3)$$

$$C_H = \frac{\kappa^2}{\ln((z_1 - d)/z_{0m}) \ln((z_1 - d)/z_{0h})}. \quad (4)$$

## 2 Governing Equations

The desingularized fast-slow system in fast time  $\tau$  is

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2}c_m \ell S^2 \tilde{e} - \frac{g}{2\theta_0} q_\theta - \frac{1}{2\ell} \tilde{e}^3, \quad (5)$$

$$\frac{dq_\theta}{d\tau} = -c_w \theta_z(T_s, \tilde{e}) \tilde{e}^3 - \frac{g}{\theta_0} c_\theta \ell q_\theta^2 - \frac{C_\theta}{\ell} \tilde{e}^2 q_\theta, \quad (6)$$

$$\frac{dS}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \mathcal{F}_S, \quad (7)$$

$$\frac{dT_s}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \frac{1}{C_s} \left( R_{\text{net}} + \rho c_p q_\theta + \frac{k_g}{d_g} (T_g - T_s) \right), \quad (8)$$

$$\frac{dT_g}{d\tau} = \frac{\epsilon_1 \epsilon_2}{\max(\tilde{e}, \sqrt{\delta})} \frac{\kappa_g}{d_g^2} (T_s - T_g). \quad (9)$$

The interior mixing length is

$$\ell(z_{\text{eff}}) = \frac{\kappa z_{\text{eff}}}{1 + \kappa z_{\text{eff}}/\ell_{\infty}}, \quad z_{\text{eff}} = z - d + z_{0m}. \quad (10)$$

### 3 Notes Traceability

The manuscript templates were aligned with project notes emphasizing:

- desingularization using  $\tilde{e} = \sqrt{e + \delta}$ ,
- geometric fast-slow interpretation of the closure manifold,
- smooth stratification regularization for Fenichel-compatible dynamics.

Primary note sources:

- `notes/Governing Equations and Turbulence Closure.md`
- `notes/Closures.md`
- `notes/Types.md`

### 4 Reproducibility

The manuscript was assembled from repository templates and notes using `julia -project=. scripts/assemble_manuscript.jl`. The primary benchmark figure is generated by `scripts/run_cases99.jl` and written to `paper/figures/cases99_manifold.png`.